Shape transitions and rare reversals in the elephant random walk

Arjun Pemmasani

Department of Mathematics, Harvey Mudd College
Claremont, CA 91711, USA
Corresponding author: apemmasani@hmc.edu
ORCID 0009-0001-2776-6298

Journal of Theoretical Probability

Review-only supplement

This supplement contains Section S6 and the bibliography of the extended manuscript. It supplies the full proofs of the threshold-certificate steps outlined in Appendix G of the journal manuscript. Read it together with `problem131_memory.pdf`.

The following pages retain their original page, equation, and statement numbers. The complete extended manuscript, `problem131_arxiv.pdf`, is also supplied as a reference file. This extract is for peer review; it is not designated for publication as supplementary information.

S6 Proofs for the threshold certificate

This section proves the lemmas that Appendix G summarizes, and so completes the proof of Theorem 4.4. We use the notation of that appendix, and in particular the inequality (9). We write $\Phi_N(t) = \sum_{n\le N} c_n (it)^n$ for the partial sums of the moment series.

S6.A One value of the memory

Lemma S6.1 (moment bounds)**.** *Let $N \ge 3$.*

(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac12 < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1+nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on S for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde A \tilde B^n}{\Gamma(1+n\alpha)}, \qquad \tilde A = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde B = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \le AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \ge n\alpha - 1 > 0$ on S, because $\alpha \le 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j| \le 1$. For $x > 0$ and real y,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k\ge0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \le \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [42, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \ge 1 + n\alpha$ and $|y| \le n\rho$. Then $y^2/x^2 \le \rho^2/\alpha^2$ and $y^2/x \le n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \ge \Gamma(1+n\alpha)$, as Γ increases on $[1.47, \infty)$. □

Lemma S6.2 (truncation)**.** *Let $0 < \alpha \le 1$, and suppose that $|c_n| \le \tilde A \tilde B^n/\Gamma(1+n\alpha)$ for $n > N$. Let $t > 0$, $0 \le k \le N$ and $n_0 = N+1$, and put*

$$u = \frac{\tilde A \tilde B^{n_0} n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)}, \qquad r = \tilde B t\Big(1 + \frac{1}{n_0}\Big)^k \frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

If $r < 1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N} c_n(it)^n\big| \le \tau_k(t) := u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde A \tilde B^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde B t(1+1/n)^k \Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma S6.3 (cells)**.** *Let ν be even, $t_c > 0$ and $|\delta| \le h$. Then $|\phi(t_c+\delta)| \le \sum_{k<\nu} |\phi^{(k)}(t_c)| h^k/k! + c_\nu h^\nu$.*

Proof. This is Taylor's formula, with $|\phi^{(\nu)}| \le \mathbb{E}|L|^\nu = \mathbb{E}L^\nu = \nu!\, c_\nu$. □

Lemma S6.4 (the tail)**.** *Let $0 < T_2 < T_3$ with $T_3 \ge 2^a T_2$, and let $\beta > 3$ with $a\beta > 1$. Cover $[0, T_3]$ by cells $[l_j, r_j]$ with $l_0 = 0$ and with T_2 among the end points, and let $s_j \ge \sup_{[l_j, r_j]} |\phi|$. Let C be the largest $s_j r_j^\beta$ over the cells that meet $[T_2, T_3]$, and put*

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \le C t^{-\beta}$ for every $t \ge T_3$.

Proof. Let $b = s_j$ on the cells in $[0, T_3)$ and $b(t) = Ct^{-\beta}$ for $t \ge T_3$. On a cell that meets $[T_2, T_3]$ we have $s_j \le C r_j^{-\beta} \le C t^{-\beta}$. So $b(t) \le Ct^{-\beta}$ for $t \ge T_2$, and $|\phi| \le b$ on $[0, T_3]$. Let $J(t) = \int_0^1 b(tv^a)\, b(t(1-v)^a)\, dv$.

We show that $J(t) \le R b(t)$ for $t \ge T_3$. By symmetry $J(t)$ is twice the integral over $v \le \frac12$. There $t(1-v)^a \ge 2^{-a} T_3 \ge T_2$, so $b(t(1-v)^a) \le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\, dv$. Put $v_0 = (T_2/t)^{1/a} \le \frac12$. On $v < v_0$ we substitute $u = tv^a$, and this part becomes

$$\frac{2}{a} t^{-1/a} \int_0^{T_2} s(u)\big(1 - (u/t)^{1/a}\big)^{-a\beta} u^{1/a - 1}\, du.$$

Here $t^{-1/a} \le \frac12 T_2^{-1/a}$ and $(u/t)^{1/a} \le \frac12 (u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1} du = a(r^{1/a} - l^{1/a})$, this part is at most the first term of R. On $v \ge v_0$ we use $b(tv^a) \le Ct^{-\beta} v^{-a\beta}$ and $(1-v)^{-a\beta} \le 2^{a\beta}$. This part is at most $2^{1+a\beta} T_2^{-\beta} (T_2/t)^{1/a} C/(a\beta - 1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau \ge T_3$ with $|\phi| \le b$ on $[0, \tau]$, and suppose that $\varsigma < \infty$. By continuity $|\phi| \le b$ on $[0, \varsigma]$. The arguments in (9) at $t = \varsigma$ lie in $[0, \varsigma]$, so $|\phi(\varsigma)| \le J(\varsigma) \le R b(\varsigma) < b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3, \infty)$, we get $|\phi| < b$ near ς, against the choice of ς. □

This is an explicit form of Step 4 in the proof of [8, Theorem 2.5], with computed constants.

Lemma S6.5 (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n \le N \\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E\Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{C T_3^{3-\beta}}{\beta - 3}.$$

Proof. As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi} \int e^{-itx} \operatorname{Re}\phi(t)\, dt$, and dominated convergence allows two derivatives under the integral [40, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\, dt$. On $[0, T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma S6.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\, dt \le \frac13 T_3^3 \tau_0(T_3)$. The integral over $t > T_3$ is at most $C T_3^{3-\beta}/(\beta - 3)$. □

S6.B Intervals of the memory

Ball arithmetic with a as an interval fails here. Near the threshold the series for ϕ cancels terms of size about 10^{60}, so the boxes in a would have to be narrower than about 10^{-65}. We expand in a around one exact point instead.

Lemma S6.6 (Cauchy remainder). *Let h be analytic near the closed disc $|z-a_0|\le\rho$, with $|h|\le M$ there and Taylor coefficients h_i at a_0. If $0<q<1$ and $|a-a_0|\le q\rho$, then $|h(a)-\sum_{i\le K}h_i(a-a_0)^i|\le Mq^{K+1}/(1-q)$.*

Proof. Cauchy's estimate gives $|h_i|\le M\rho^{-i}$. Sum the geometric tail. □

Let $I=[a_0-w,a_0+w]$, $0<q<1$ and $\rho=w/q$, and let S be the square $a_0+[-\rho,\rho]+i[-\rho,\rho]$, which contains the disc of Lemma S6.6. Suppose that $\alpha=a_0-\rho>\frac12$, that $|z|\le1$ on S, and that $\tilde A,\tilde B$ are as in Lemma S6.1(b). Let τ_k be the bound of Lemma S6.2 with these constants. It holds for every real $a\in I$. For a cell with centre t_c write $\partial_t^k\Phi_N(z,t_c)=E_k(z)+iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k=\sum_{n\le N}\sup_S|c_n|\,n(n-1)\cdots(n-k+1)\,t_c^{n-k}$.

Proposition S6.7 (enclosure on an interval). *(a) For every $a\in I$ and every t in the cell with centre t_c and half-width h,*

$$|\phi_a(t)|\le\sum_{k<12}\Big(\sum_{i\le K}w^i\sqrt{E_{k,i}^2+O_{k,i}^2}+\frac{M_kq^{K+1}}{1-q}+\tau_k(t_c)\Big)\frac{h^k}{k!}+h^{12}\sup_S|c_{12}|.$$

(b) Let T_2, $T_3\ge2^{a_0+w}T_2$, β and the cells be as in Lemma S6.4, with s_j the bound of (a), and suppose that $R<1$ for every $a\in I$. Let $F(z)=\sum_{n\le N,\,n\ even}(-1)^{n/2}c_n(z)T_3^{n+3}/(n+3)$, with Taylor coefficients F_i at a_0, and let M_F be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a\in I$

$$\Big|g_a''(0)+\frac1\pi\sum_{i\le K}F_i(a-a_0)^i\Big|\le\eta=\frac1\pi\Big(\frac{M_Fq^{K+1}}{1-q}+\frac{T_3^3}{3}\tau_0(T_3)+\frac{CT_3^{3-\beta}}{\beta-3}\Big),$$

and $a\mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma S6.3 with $\nu=12$, and Lemma S6.2 to replace $\phi_a^{(k)}$ by $\partial_t^k\Phi_N$. For real a we have $|\partial_t^k\Phi_N(a,t_c)|=|(E_k(a),O_k(a))|$. Lemma S6.6 for E_k and O_k, with $\sup|E_k|+\sup|O_k|\le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a)\le\sup_S|c_{12}|$.

For (b), $T_3\ge2^aT_2$ for every $a\in I$, so Lemma S6.4 applies at each a. Then Lemma S6.5 and Lemma S6.6 for F give the enclosure. By Lemma S6.1(b) the series for $\phi_a(t)$ converges uniformly for $a\in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0,T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a\in I$. Dominated convergence gives the continuity. □

Near $a=\frac12$ the bounds of Lemma S6.1 degenerate, since $m_2=a/(2a-1)$. We rescale.

Lemma S6.8 (the scaled variable). *Put $y=2a-1$ and $\mu_n=m_ny^{\lfloor n/2\rfloor}$.*

(a) We have $\mu_0=\mu_1=1$, $\mu_2=a$, and for $n\ge3$

$$(na-\theta_n)\mu_n=\sum_{j=1}^{n-1}\theta_jy^{e_{nj}}\mu_j\mu_{n-j},$$

where $e_{nj}=1$ if n is even and j is odd, and $e_{nj}=0$ otherwise. Here $na-\theta_n$ is $(n-1)a$ for odd n and $na-1$ for even n. So every μ_n is a rational function of a with no pole in $\operatorname{Re}a>\frac14$.

(b) Let $\sigma^2 = \mathbb{E}L^2$, $X = L/\sigma$ *and* $\hat{c}_n = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1+an)$. *Then* $\sigma^2 = 1/(y\Gamma(2a))$,

$$\mathbb{E}e^{iuX} = \sum_{n \text{ even}} \hat{c}_n (iu)^n + \sqrt{y\Gamma(2a)} \sum_{n \text{ odd}} \hat{c}_n (iu)^n, \qquad g_a''(0) = -\frac{1}{\pi\sigma^3} \int_0^\infty u^2 \operatorname{Re} \mathbb{E}e^{iuX}\, du.$$

Proof. Multiply the recursion for m_n by $y^{\lfloor n/2 \rfloor}$. The exponent $\lfloor n/2 \rfloor - \lfloor j/2 \rfloor - \lfloor (n-j)/2 \rfloor$ is 1 when n is even and j is odd, and 0 otherwise. Also $m_2 = a/y$. This proves (a), as the zeros of the factors are $a = 0$ and $a = 1/n \le \frac{1}{4}$. For (b), $\sigma^2 = 2m_2/\Gamma(1+2a) = 1/(y\Gamma(2a))$ and $\mathbb{E}X^n/n! = c_n \sigma^{-n} = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor} (y\Gamma(2a))^{(n \bmod 2)/2}/\Gamma(1+an)$. The formula for $g_a''(0)$ follows from the substitution $t = u/\sigma$ in Lemma S6.5. □

The fixed-point equation is linear, so X satisfies it too, and (9) and Lemmas S6.3 to S6.5 hold for X. The proof of Lemma S6.1(b) carries over to μ_n for $n \ge 3$ and $\alpha > \frac{1}{4}$, with two changes. The constant A becomes $((N+1)\alpha - 1)/(N\kappa)$, where κ is the largest value of $\max(1, |z|) \max(1, |2z-1|)$ on S. And $\tilde{B}$ gains the factor $\max(1, \sup_S |\Gamma(2z)|)^{1/2}$. For real a the odd part carries the factor $\sqrt{y\Gamma(2a)}$. In the cell bounds we treat the even and odd parts as separate analytic functions. We multiply the odd part by $f = \sup_I \sqrt{y\Gamma(2a)}$ only for real a, and the remainders by $\max(1, f)$. The real part of $\mathbb{E}e^{iuX}$ involves only the even part. So the disc may cross $a = \frac{1}{2}$, and the cover reaches $a = \frac{1}{2}$. The sign of $g_a''(0)$ is the opposite of the sign of the integral, since $\sigma^3 > 0$ and the Fourier formula has a minus sign.

S6.C From the runs to the theorem

The script accepts an interval of a only when $G \pm \eta$ has one sign on each piece of a subdivision of it. In the 7 scaled runs every piece of every interval is negative, so $g_a''(0) < 0$ on $(\frac{1}{2}, 0.668]$. The zero run has 8 intervals that cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ every piece is negative. On the 5 intervals that cover $[0.6692, 0.6712]$ every piece is positive. On $[0.6688, 0.6692]$ the pieces up to z_- are negative, the pieces from z_+ on are positive, and only (z_-, z_+) is undecided. Proposition S6.7(b) turns these signs into $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The order matters. A positive interval left of z_- would give another zero, and parts (a) and (c) of Theorem 4.4 would fail. So the script prints the sign of each accepted interval with its exact end points. Its mode `zero` fails unless the signs are $-$ before the undecided piece and $+$ after it. Its mode `check` requires the same of the stored records, together with the sign $-$ on every interval of the scaled runs.

References

[1] G. M. Schütz and S. Trimper, *Elephants can always remember: exact long-range memory effects in a non-Markovian random walk*, Phys. Rev. E **70** (2004), 045101(R). https://doi.org/10.1103/PhysRevE.70.045101.

[2] L. Addario-Berry, L. Devroye, G. Lugosi and V. Velona, *Broadcasting on random recursive trees*, Ann. Appl. Probab. **32** (2022), 497–528. https://doi.org/10.1214/21-AAP1686.

[3] E. Althaus, L. Hartung and R. Steiner, *A random walk approach to broadcasting on random recursive trees*, J. Theor. Probab. **39** (2026), article no. 65. https://doi.org/10.1007/s10959-026-01519-z.

[4] C. F. Coletti, R. Gava and G. M. Schütz, *Central limit theorem and related results for the elephant random walk*, J. Math. Phys. **58** (2017), 053303. https://doi.org/10.1063/1.4983566.

[5] B. Bercu, *A martingale approach for the elephant random walk*, J. Phys. A: Math. Theor. **51** (2018), 015201. https://doi.org/10.1088/1751-8121/aa95a6.

[6] N. Kubota and M. Takei, *Gaussian fluctuation for superdiffusive elephant random walks*, J. Stat. Phys. **177** (2019), 1157–1171. https://doi.org/10.1007/s10955-019-02414-0.

[7] H. Guérin, L. Laulin, K. Raschel and T. Simon, *On the limit law of the superdiffusive elephant random walk*, Electron. J. Probab. **30** (2025), paper no. 102, 1–25. https://doi.org/10.1214/25-EJP1362.

[8] H. Guérin, L. Laulin and K. Raschel, *A fixed-point equation approach for the superdiffusive elephant random walk*, Ann. Inst. Henri Poincaré Probab. Stat. **62** (2026), 973–1005. https://doi.org/10.1214/24-AIHP1534.

[9] H. Trinas and G. Valle, *A superdiffusive local limit theorem for the elephant random walk and the breakdown of log-concavity*, preprint, 2026. arXiv:2609.14753.

[10] H. Murahara and G. Tokumitsu, *Optimal rates in local and central limit theorems for the elephant random walk*, preprint, 2026. arXiv:2610.35019.

[11] P. Vishwakarma, *Moments approach for the elephant random walk*, preprint, 2026. arXiv:2609.11035.

[12] P. Kagey, *Problem 131*, Open Problem Collection, https://peterkagey.com/problems/131/ (accessed October 2026).

[13] A. Pemmasani, *Uniform Bessel bounds for magnetization crossings in finite Ising chains*, preprint, 2026. arXiv:2610.06935.

[14] X. Fan and Y. Qi, *A generalized elephant random walk with time-dependent memory*, preprint, 2026. arXiv:2610.16065.

[15] R. Roy, M. Takei and H. Tanemura, *The elephant random walk in the triangular array setting*, J. Appl. Probab. **62** (2025), 997–1009. https://doi.org/10.1017/jpr.2024.106.

[16] R. Kürsten, *Random recursive trees and the elephant random walk*, Phys. Rev. E **93** (2016), 032111. https://doi.org/10.1103/PhysRevE.93.032111.

[17] E. Baur, *Percolation on random recursive trees*, Random Struct. Algorithms **48** (2016), 655–680. https://doi.org/10.1002/rsa.20603.

[18] J. Engländer and S. Volkov, *Turning a coin over instead of tossing it*, J. Theor. Probab. **31** (2018), 1097–1118. https://doi.org/10.1007/s10959-016-0725-1.

[19] E. Baur and J. Bertoin, *Elephant random walks and their connection to Pólya-type urns*, Phys. Rev. E **94** (2016), 052134. https://doi.org/10.1103/PhysRevE.94.052134.

[20] J. Bertoin, *Sizes of the largest clusters for supercritical percolation on random recursive trees*, Random Struct. Algorithms **44** (2014), 29–44. https://doi.org/10.1002/rsa.20448.

[21] T. Miyazaki and M. Takei, *Limit theorems for the 'laziest' minimal random walk model of elephant type*, J. Stat. Phys. **181** (2020), 587–602. https://doi.org/10.1007/s10955-020-02590-4.

[22] D. A. Kessler and H. Levine, *Scaling solution in the large population limit of the general asymmetric stochastic Luria–Delbrück evolution process*, J. Stat. Phys. **158** (2015), 783–805. https://doi.org/10.1007/s10955-014-1143-3.

[23] D. A. Kessler and H. Levine, *Large population solution of the stochastic Luria–Delbrück evolution model*, Proc. Natl. Acad. Sci. USA **110** (2013), 11682–11687. https://doi.org/10.1073/pnas.1309667110.

[24] P. Bokes, A. Hlubinová and A. Singh, *Reversible transitions in a fluctuation assay modify the tail of Luria–Delbrück distribution*, Axioms **12** (2023), 249. https://doi.org/10.3390/axioms12030249.

[25] A. G. Pakes, *Remarks on the Luria–Delbrück distribution*, J. Appl. Probab. **30** (1993), 991–994. https://doi.org/10.2307/3214530.

[26] M. Möhle, *Convergence results for compound Poisson distributions and applications to the standard Luria–Delbrück distribution*, J. Appl. Probab. **42** (2005), 620–631. https://doi.org/10.1239/jap/1127322016.

[27] J. Schweinsberg, *Dynamics of the evolving Bolthausen–Sznitman coalescent*, Electron. J. Probab. **17** (2012), paper no. 91, 1–50. https://doi.org/10.1214/EJP.v17-2378.

[28] J. Bertoin, *On the non-Gaussian fluctuations of the giant cluster for percolation on random recursive trees*, Electron. J. Probab. **19** (2014), paper no. 24, 1–15. https://doi.org/10.1214/EJP.v19-2822.

[29] A. Iksanov and M. Möhle, *A probabilistic proof of a weak limit law for the number of cuts needed to isolate the root of a random recursive tree*, Electron. Commun. Probab. **12** (2007), 28–35. https://doi.org/10.1214/ECP.v12-1253.

[30] A. Meir and J. W. Moon, *Cutting down recursive trees*, Math. Biosci. **21** (1974), 173–181. https://doi.org/10.1016/0025-5564(74)90013-3.

[31] C. Desmarais, C. Holmgren and S. Wagner, *Broadcasting-induced colorings of preferential attachment trees*, Random Struct. Algorithms **63** (2023), 364–405. https://doi.org/10.1002/rsa.21142.

[32] T. Kardas, V. Shneer and V. Wachtel, *Broadcasting on random trees: martingale methods and error bounds*, preprint, 2026. arXiv:2606.22960.

[33] A. Gut and U. Stadtmüller, *Bernoulli elephant random walks*, J. Theor. Probab. **39** (2026), article no. 60. https://doi.org/10.1007/s10959-026-01517-1.

[34] K. S. Kölbig and B. Schorr, *A program package for the Landau distribution*, Comput. Phys. Commun. **31** (1984), 97–111. https://doi.org/10.1016/0010-4655(84)90085-7.

[35] M. Yamazato, *Unimodality of infinitely divisible distribution functions of class L*, Ann. Probab. **6** (1978), 523–531. https://doi.org/10.1214/aop/1176995474.

[36] F. Johansson, *Numerical integration in arbitrary-precision ball arithmetic*, in: Mathematical Software – ICMS 2018, Lect. Notes Comput. Sci. **10931**, Springer, 2018, 255–263. https://doi.org/10.1007/978-3-319-96418-8_30.

[37] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292. https://doi.org/10.1109/TC.2017.2690633.

[38] S. Dharmadhikari and K. Joag-dev, *Unimodality, Convexity, and Applications*, Academic Press, Boston, 1988.

[39] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, Cambridge, 2004. https://doi.org/10.1017/CBO9781139165372.

[40] R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge University Press, Cambridge, 2019. https://doi.org/10.1017/9781108591034.

[41] H. Scheffé, *A useful convergence theorem for probability distributions*, Ann. Math. Stat. **18** (1947), 434–438. https://doi.org/10.1214/aoms/1177730390.

[42] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.8 of 2026-09-15, https://dlmf.nist.gov.

[43] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, J. Am. Stat. Assoc. **58** (1963), 13–30. https://doi.org/10.1080/01621459.1963.10500830.